

MINI SENTRIX SOC — ARCHITECTURE SUMMARY

Final Demo Day | Judge Reference Sheet

Team: 5 Members | Duration: 4 Days | Stack: 100% Open Source & Local

1. NETWORK & ENVIRONMENT

Component	Details
Network Type	Host-Only / Internal — zero external internet dependency
Attacker VM	Kali Linux — isolated on host-only network
Target VM	Metasploitable2 — 192.168.100.20
Security VM	Ubuntu 24.04.4 LTS (root) — 10.9.26.170 — runs Suricata IDS + Wazuh Agent
WAF Host	Vansh-laptop (Windows 11) — 10.9.15.118 — Docker: ModSecurity + Nginx (port 80→8080)
Agent: Peter_Parker	Windows 11 — 10.9.210.4 — monitored endpoint
Wazuh Manager	wazuh.manager — Docker on Mac — collects from all 4 agents

2. THE 4 MANDATORY LAYERS

Layer	Tool	Function	Why This Tool
Perimeter & Detection	Suricata IDS	Detects malicious traffic, generates real-time alerts	AF_PACKET mode on virtual interface — no promiscuous mode required; SRM-compatible
SIEM & Visibility	Wazuh Manager + Wazuh Dashboard (OpenSearch)	Collects logs from all 4 agents, live 5-panel dashboard	Docker Compose deployable, no external connectivity, built-in correlation engine
Identity & Access	Keycloak (SSO + RBAC)	Protects internal service with authentication; admin + analyst roles defined	Fully local Docker deployment, industry-standard OAuth2/OIDC, MFA-capable
Automated Response	Shuffle SOAR + iptables	Playbook triggers on Wazuh alert → blocks attacker IP via firewall rule, zero human steps	Docker Compose, visual workflow builder, direct iptables integration

3. ATTACK → DETECTION → RESPONSE DATA FLOW

Kali Linux (Attack) → Suricata IDS (Alert) → Wazuh Agent → Wazuh Manager → Wazuh Dashboard (SIEM)

WAF (ModSecurity block) → modsec_audit.log → Wazuh Agent (Vansh-laptop) → Wazuh Manager → Dashboard (separate source)

Wazuh Alert → Shuffle Trigger → iptables block rule → Attacker IP blocked → Dashboard updated

4. SIEM DASHBOARD — 5 PANELS

Panel	What It Shows	Data Source
Alert Count by Type	Bar chart — event counts grouped by rule.description	Wazuh (all agents)
Top Source IPs	Bar chart — highest-volume source IPs by alert count	Wazuh (agent.ip field)
WAF Blocks	ModSecurity blocked + rejected events over time	Wazuh — Vansh-laptop agent (WAF source)
Authentication Events	Keycloak login events, rule.groups: authentication	Wazuh — Keycloak/auth rules
Critical Security Events	Donut chart — rule.level 7 vs 10 distribution	Wazuh (all agents)

5. ATTACK SIMULATIONS & EXPECTED RESULTS

Attack	Tool	Expected Alert	Expected Action
Port Scan	nmap -sS -T4 192.168.100.20	Suricata: ET INFO Possible Kali Linux host scan	Alert in Wazuh, visible in dashboard
Brute Force	Hydra against SSH / login	Suricata brute force + Wazuh auth failure rule	Shuffle triggers → iptables blocks attacker IP
SQL Injection	SQLMap against DVWA	WAF: ModSecurity blocked malicious request (rule 100100)	403 response + block event in Wazuh WAF panel

6. Brownie Points Implemented (Day 3 & Day 4)

Day 3 — "The Alarm That Cried Wolf"

Riddle Decode: Every alert carries equal weight regardless of context. The system screamed 47 times — by alert 31, no one responded. An attacker who studied this pattern knew exactly when to strike. The fix: alert prioritization based on cumulative context, not individual events.

What We Built:

- Configured Wazuh rule levels to differentiate low-noise events (level 3) from critical detections (level 7, 10)
- Dashboard Critical Security Events panel shows level distribution — analysts see instantly which alerts need action
- Demonstrated: Hydra brute force escalates to level 10 while routine scans stay at level 3 — same source, different weight based on behavior pattern

Day 4 — "The Map That Only Shows Roads You Have Already Walked"

Riddle Decode: Signature-based detection only catches known attacks. An attacker who takes a novel or slow path — one with no existing signature — is invisible to the system. "Small enough that no single step looked like a journey" = low-and-slow attacks that fall below every threshold individually.

What We Built:

- Anomaly detection layer using Wazuh frequency rules — if a source IP generates $3\times$ its rolling average of failed events in a 5-minute window, it escalates regardless of whether it matches a known signature
- Demonstrated: modified slow Hydra attack (reduced thread count, random delays) — no single event matched the Hydra signature, but the volume anomaly triggered a CRITICAL alert
- The mathematical framing: detection shifts from pattern matching to distribution analysis

7. TEAM ROLES

Member	Role	Owns
Member 1 (Aabidha)	Infrastructure & Stability Lead	VM setup, host-only network, architecture diagram, iptables
Member 2 (Heyden)	IDS / Detection Engineer	Suricata rules, attack simulation execution, alert verification
Member 3 (Amal)	SIEM & Dashboard Engineer	Wazuh Manager, 5-panel dashboard, log source verification, this document
Member 4 (Vansh)	WAF & Web Security Engineer	ModSecurity + Nginx, DVWA, WAF log forwarding to Wazuh
Member 5 (Goutham)	IAM & SOAR Engineer	Keycloak SSO/RBAC, Shuffle playbooks, automated IP block pipeline
